NSR/AS Lab 3 – VPNs, Pattarapol Tantechasa, 103883220

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***Abstract*— This report introduces VPNs, with both theoretical background and hands-on configuration. The first part outlined theory of VPNs, their characteristics, capabilities and how it can be used to implement organizational security policy. Furthermore, practical lab activities to dive deeper into how to set up a simple encrypted VPN between the two Ubuntu machines using Linux VPN called OpenVPN, observe and analyze traffic behavior using Wireshark. Results confirmed that data is encrypted on the ethernet interface and decrypted within the tunnel. The lab reinforces key networking principles, demonstrates how VPN works and builds foundational knowledge in secure network communication.**

Keywords—VPN, Tunnelling, OpenVPN, IP Address, Telnet, Encryption, Network Security.

I. Introduction to VPNs

A Virtual Private Network (VPN) allows users to establish a protected network connection over public networks [1]. It enables secure transmission of sensitive data across untrusted networks such as the internet [2]. VPNs connect two devices through an encrypted tunnel, securing packet communication and masking IP addresses from external observers.

A. VPNs Characteristics and Capabilities

1) Tunnelling: VPNs rely on tunnelling protocols, this encapsulate data inside a secure communication channel. This allows different network layers to be used depending on the protocol. For example, IP Security (IPSec) uses Network Layer protocol (Layer 3) [3].

2) Encryption: Another core characteristics of a VPN, it ensures confidentiality even when data is transmitted across insecure public networks like internet [3].

3) Data Integrity: VPNs ensure that data transmitted through tunnel can't be altered by hash-based algorithms such as HMAC. Integrity is handled by The Authentication Header (AH) for authentication and The Encapsulating Security Payload (ESP) to provide both encryption and integrity [2], [4].

4) Authentication: VPNs authenticate users before establishing a connection, using method like CHAP or EAP [3], [5].

5) Remote Access Connectivity: VPNs allow mobile users and remote workers to securely access internal organizational resources as if they were on-site [3].

6) Cost Effective: Instead of using expensive leased lines from Internet Service Provider (ISP), organizations can use VPNs over public internet infrastructure to interconnect branch offices [3].

7) Controlled Third-Party Access: VPNs enable secure, selective access for partners, clients or contractors to organization sensitive information like stock on hand [3].

8) Scalability and Flexibility: VPNs support a large number of users and can scale effectively across various sizes of organisation from small to enterprises without significant infrastructure changes. Also, support multiple VPN types, which makes it flexible for different business needs [3].

B. How VPNs can be used to implement organisational security policy

VPNs play a vital role during implementation of security policy for an organization, as they enable secure remote access, maintaining data confidentiality and integrity, and controlling user access to sensitive resources [3]. Organization can ensure that only authorized users can access internal systems through authentication mechanisms such as CHAP, thereby enforcing access control policies [5]. Encryption protocols like IPSec are used to protect data in transit and help to hide source IP addresses by encapsulating traffic within secure tunnels, aligning with organizational requirements for confidentiality and integrity [4]. As well as, VPNs enforce authorization policies, allow control over which users or partners can access certain resources, from which locations and at what time [3]. As a result, VPNs are not only a tool for connectivity but also a critical enabler for applying and maintaining organizational cybersecurity policies [3].

II. OpenVPN Behaviour

This lab conducted a simple encrypted VPN between the two Ubuntu machines using Linux VPN software called OpenVPN. First, I began with setting up two Ubuntu machines and check IP Address of both machine with "*ifconfig*" command, as shown in Table I. As well as check connection between the two machines with "*ping*" command.

Table I. Initiail Virtual Machines IP Address

Virtual Machine	IP Address
Ubuntu 1	192.168.120.128
Ubuntu 2	192.168.120.129

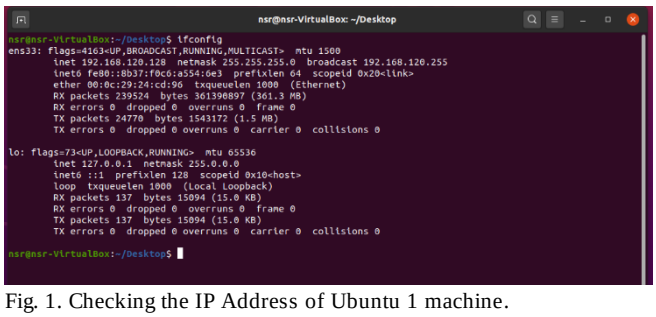

Fig. 1. Checking the IP Address of Ubuntu 1 machine.

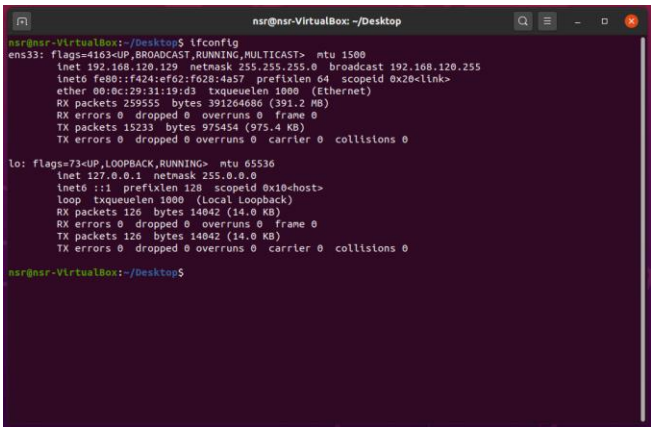

Fig. 2. Checking the IP Address of Ubuntu 2 machine.

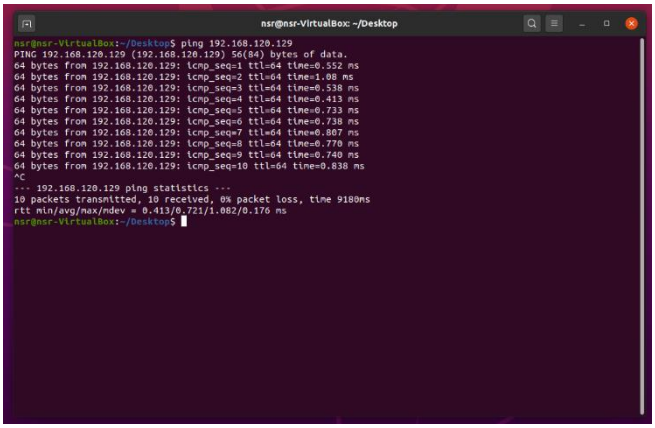

Fig. 3. Ping to check connection between the two machines.

After connectivity is established, I generated the shared password named *mykey* on Ubuntu 1 machine with "*openvpn –genkey –secret mykey*" command. Verify that key has been created with "*ls -l*" command and examine the file with "*cat mykey*" command.

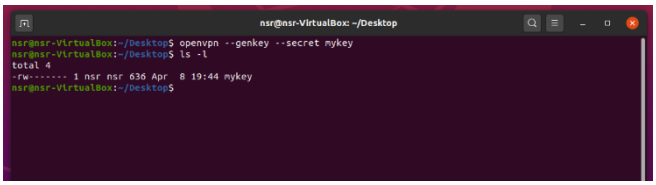

Fig. 4. Generate and verify "*mykey*" on Ubuntu 1 machine.

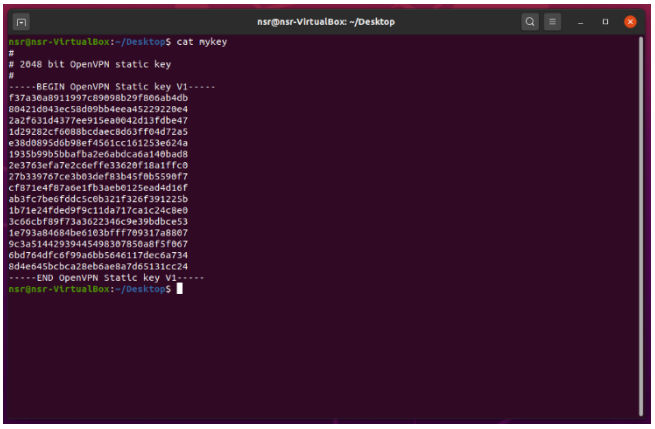

Fig. 5. Examine "*mykey*" on Ubuntu 1 machine.

Now that the shared key has been successfully generated, I copy and transfer *mykey* to Ubuntu 2 machine with Secure Copy (*scp*). Firstly, I installed and updated ssh (includes *scp*) on both machines with following command: "*sudo apt update*", "*sudo apt-get install openssh-server*".

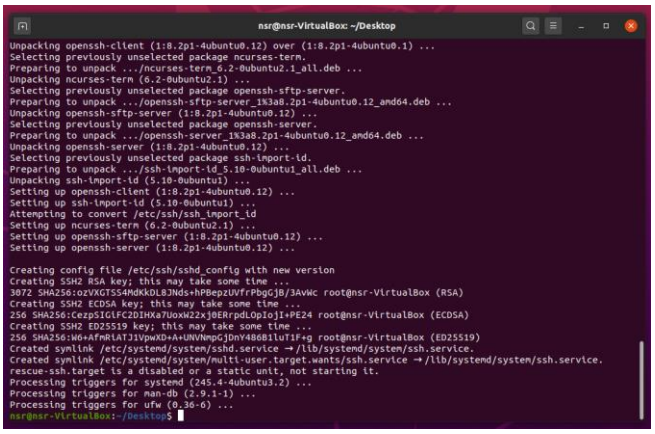

Fig. 6. Installed and Updated ssh on Ubuntu 1.

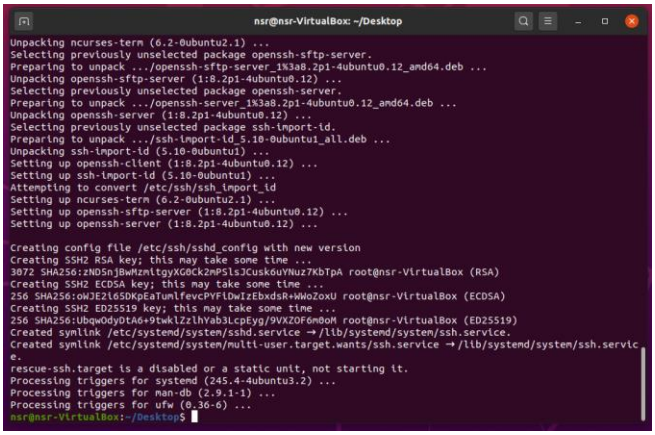

Fig. 7. Installed and Updated ssh on Ubuntu 2.

Now back to Ubuntu 1 machine, I run the command "*scp mykey nsr@192.168.120.129:Desktop/mykey*" to copy and transfer the shared key to Ubuntu 2 machine.

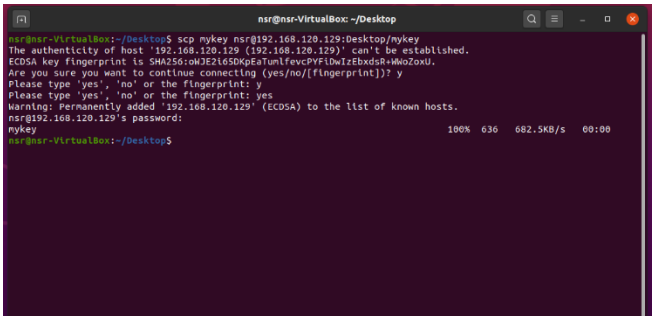

Fig. 8. Copy and Transfer *mykey* from Ubuntu 1 to Ubuntu 2 machine.

Ensuring *mykey* has been successfully transfer by run the command "*ls -l*" and "*cat mykey*" on Ubuntu 2 machine.

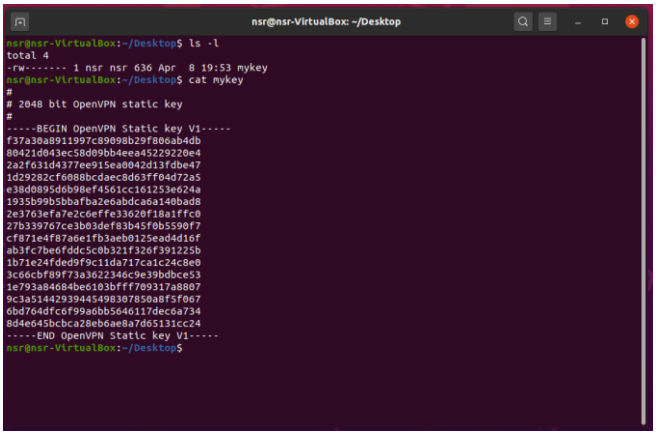

Fig. 9. Verify successful transfer *mykey* on Ubuntu 2 machine.

TABLE II. VIRTUAL MACHINES IP ADDRESS AND TUNNEL END POINTS

Virtual Machine	IP Address	Tunnel End Points
Ubuntu 1	192.168.120.128	10.4.0.1
Ubuntu 2	192.168.120.129	10.4.0.2

Now I have begun the process of tunnel between the two machines, as shown in Table II. Setup tunnel on Ubuntu 1 machine by run the command "*sudo openvpn –remote 192.168.120.129 – dev tun1 –ifconfig 10.4.0.1 10.4.0.2 –verb 5 –secret mykey*".

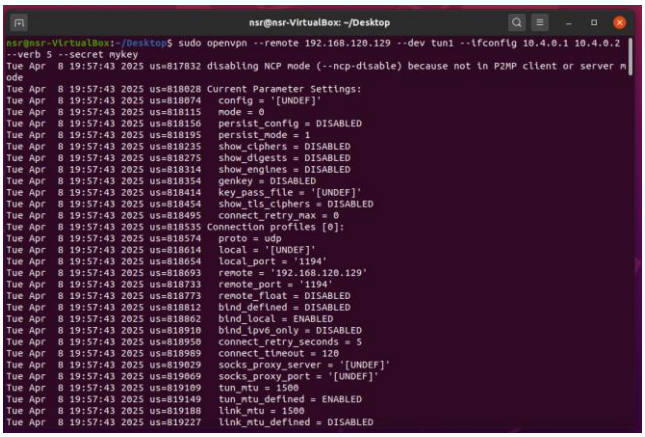

Fig. 10. Setup tunnel on Ubuntu 1 machine.

On Ubuntu 2, set up the tunnel in the reverse direction so that traffic can travel in both directions and verification.

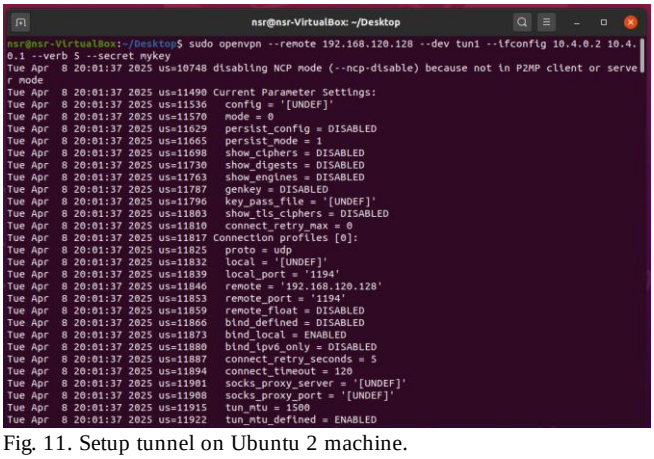

Fig. 11. Setup tunnel on Ubuntu 2 machine.

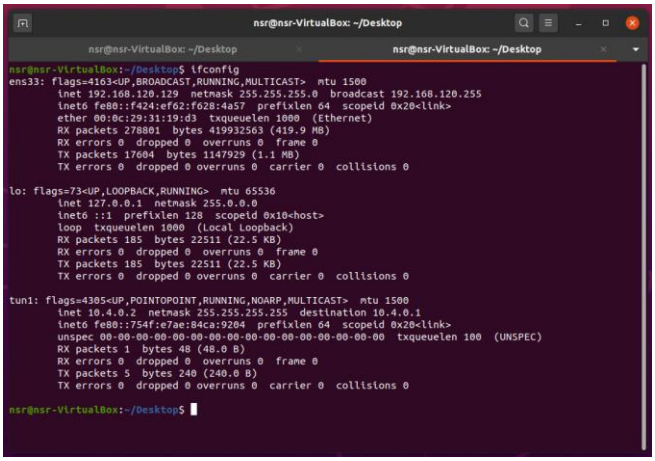

Fig. 12. Verify successful tunnel on Ubuntu 2 machine.

After successfully set up tunnel between the two machines, I ping Ubuntu 1 from Ubuntu 2 using tunnel end point of Ubuntu 1, command as follows: "*ping 10.4.0.1*". This way both machines communicate to each other through tunnel just set up.

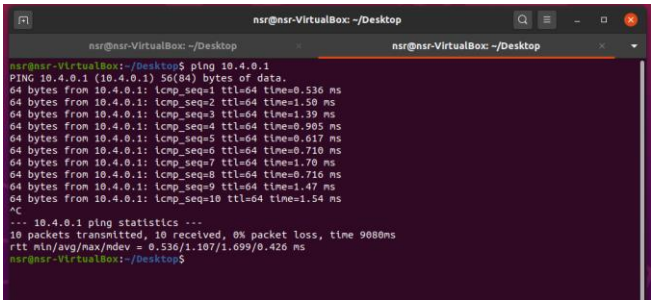

Fig. 13. Ping Ubuntu 1 from Ubuntu 2 through tunnel.

Lastly, I observed traffic between the two-machine through tun1 and *ens33* interfaces captured on Wireshark to answer questions below.

A. Is the traffic encrypted? Why?

When observing traffic on the tunnel interface (*tun1*) using Wireshark, it appears that the traffic is not encrypted. This is expected because the tun1 interface represents the internal end of the VPN tunnel. We can see IP addresses of the tunnel endpoints and transport protocols. Encryption is meant to protect data in transit over the public network or from external observers. When we observe from the inside of the tunnel, the VPN software has already decrypted the packets, which is why the traffic is readable.

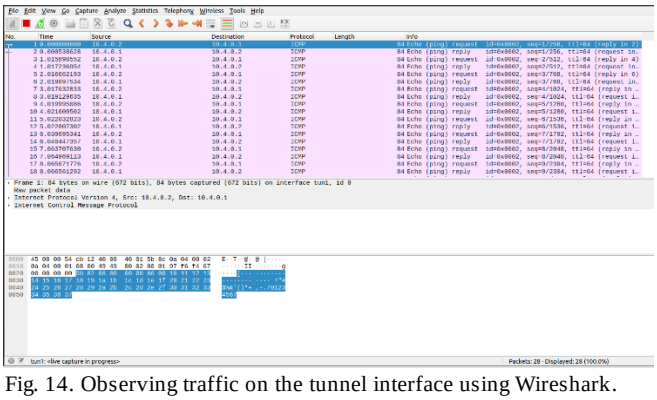

Fig. 14. Observing traffic on the tunnel interface using Wireshark.

B. *Is the traffic encrypted when monitoring the ethernet interface?*

Yes, the traffic is encrypted when observing the Ethernet interface (*ens33*). As we can see that Wireshark can't interpret the contents of the packets and flags them as "Malformed Packet" due to encryption.

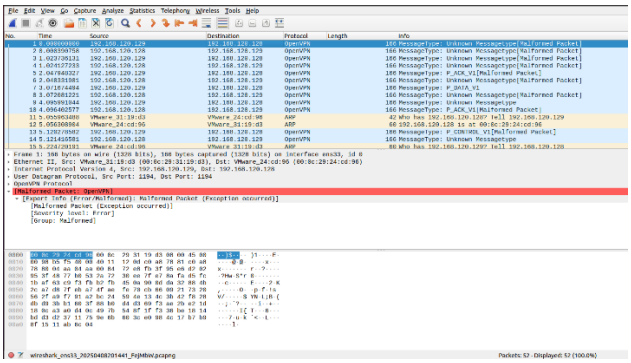

Fig. 15. Observing traffic on the ethernet interface using Wireshark.

C. *How do you explain what you see?*

We can see on the ethernet interface (*ens33*) is an encrypted VPN traffic inside an encapsulating protocol (*OpenVPN*). While the source and destination IP addresses of the VPN tunnel endpoints must remain visible for routing purposes, the payload itself is encrypted. Therefore, Wireshark shows OpenVPN packets using UDP port 1194 but cannot interpret their contents, which confirms that confidentiality is maintained.

D. *On which interface is traffic encrypted and which interface not encrypted? Why?*

I telnet from Ubuntu 2 to Ubuntu 1 with "*telnet 10.4.0.1*" command, then ensure successful logged into the remote machine with "*ifconfig*" command. Observing the tunnel interface (tun1) for traffic after running some simple command like "*ifconfig*" and "*ls*".

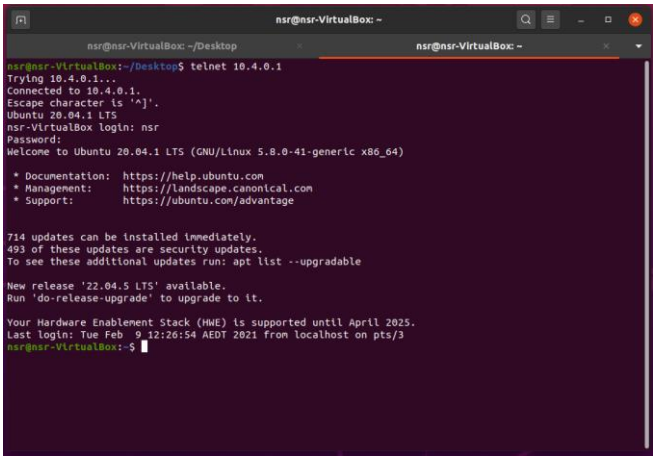

Fig. 16. Telnet from Ubuntu 2 to Ubuntu 1 through tunnel endpoints.

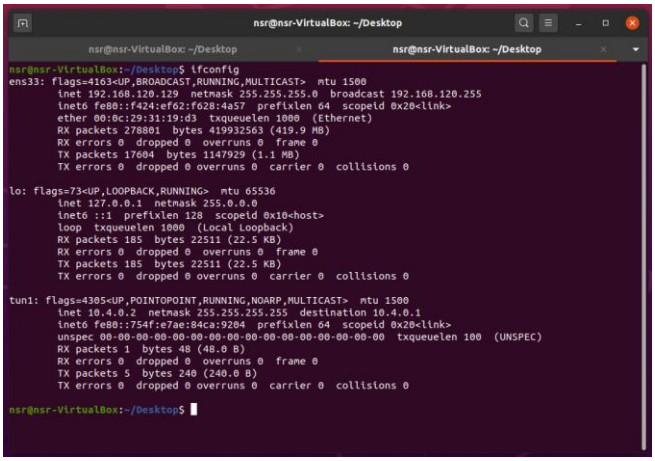

Fig. 17. Verify successfully logged into Ubuntu 1 through Telnet.

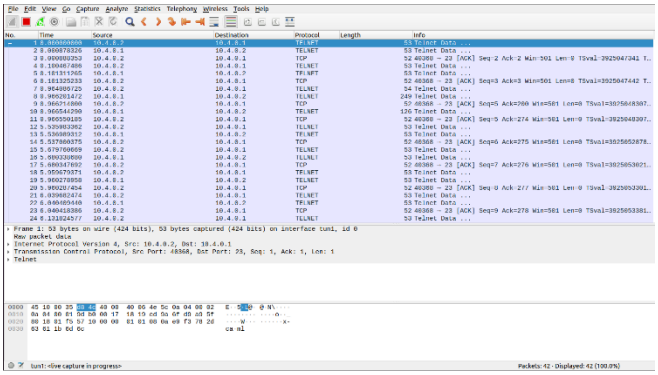

Fig. 18. Observing traffic on the tunnel interface (*tun1*).

Traffic on the ethernet interface (*ens33*) is encrypted because this interface handles the transmission of VPN packets over the network. These packets are encapsulated using OpenVPN over UDP and the original data, including Telnet sessions, is protected within the encrypted payload.

Unlike the tunnel interface (*tun1*), traffic is not encrypted because this interface represents the internal side of the VPN tunnel, where's data has already been decrypted by the VPN software, as shown in Figure 18.

III. Conclusion

In this lab, I successfully set up a simple encrypted VPN between the two Ubuntu machines using OpenVPN and verified its functionality through both command-line tools and packet inspection. Observing traffic using Wireshark on different interfaces, I was able to observe the behavior of encrypted and unencrypted traffic as it flows through and outside the VPN tunnel.

Traffic captured on the ethernet interface (*ens33*) appeared encrypted and unreadable, confirming that OpenVPN was securely encapsulating the data. Although, traffic captured on the tunnel interface (*tun1*) was visible and readable, such as ICMP (*ping*) and Telnet data.

In conclusion, I gained practical understanding of key VPN concepts, including tunneling, encapsulation and encryption. This hands-on experience deepened my understanding of how VPNs implement confidentiality, integrity and secure remote access, which are essential components of modern network security

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